Individual differences in submission timeliness in technology-enhanced learning: A growth mixture modeling approach

Sami Heikkinen[a], Henrik Bellhäuser[b], Esther van der Stappen[c], Mohammed Saqr[d]

[a] LAB University of Applied Sciences, Mukkulankatu 19, Lahti, Finland

[b] Johannes Gutenberg University Mainz, Binger Str. 14-16, Mainz, Germany

[c] Avans University of Applied Sciences, Research Group Digital Education, P.O.Box 90.116, 4800RA, Breda, the Netherlands

[d] The University of Eastern Finland, School of Computing, Yliopistonkatu 2, 80100, Joensuu, Finland

Corresponding author: Sami Heikkinen, sami.heikkinen@lab.fi

Acknowledgements

We thank the students who took part in the course and agreed to the use of their course data for research.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT author statement

Sami Heikkinen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing: original draft. **Henrik Bellhäuser:** Conceptualization, Methodology, Writing: review and editing. **Esther van der Stappen:** Conceptualization, Writing: review and editing. **Mohammed Saqr:** Methodology, Validation, Writing: review and editing.

Abstract

Students who set their own deadlines differ in whether they keep them, but aggregate lateness rates hide how these differences unfold over a course. In an asynchronous higher education course (N = 262), we modeled submission timeliness against self-set deadlines across up to eleven assignments with growth mixture modeling. Four classes emerged: On track (47%), Persistently late (10%), Escalating delay (21%) and Withdrawing (22%). Sequence analysis showed that the two late-submitting classes differed qualitatively: one was stably late, the other increasingly unstable. The classes hardly differed in the target grades students set at course start ($\varepsilon^2 = .03$) and not in the completeness or coherence of their plans. Students thus aimed and planned alike but differed in enacting their plans, which calls for early support matched to the type of delay.

Educational relevance and implications statement

Students who plan their own study schedules aim for similar grades and write similar plans, yet they follow very different paths in keeping them. About half stay ahead of their own deadlines throughout the course, one in ten are reliably late but still finish their work, and the rest either slip further behind over time or stop submitting at all early in the course. These paths already look different at the first few deadlines of the course. Teachers can use this by offering support early and matching it to the type of delay, instead of sending the same reminder to every late student.

Keywords: Self-regulated learning; Procrastination; Learning analytics; Growth mixture modeling; Sequence analysis

1 Introduction

Higher education increasingly offers flexible, asynchronous courses in which students decide when and how they study. This flexibility serves students who combine their studies with work and family, but it moves the responsibility for pacing from the course to the student. Research on self-regulated learning (SRL) consistently identifies time management as one of the regulatory strategies most closely related to achievement in online settings (Broadbent, 2017; Broadbent & Poon, 2015), and reviews of online and open learning show that these environments place particular demands on students' ability to plan and monitor their own work (Wong et al., 2019). Late and missing submissions are among the most visible signs that this regulation has broken down. They are also costly: unfinished work accumulates, feedback arrives too late to be useful, and courses are left incomplete.

Many flexible courses ask students to structure their own schedule, for instance by setting personal deadlines for their assignments. When the student has set the deadline, lateness is not a breach of an external rule but a failure to keep a commitment made to oneself. The distinction matters because external structure alone does not solve the problem of delay: course deadlines have been found not to prevent students from cramming (Theobald et al., 2021), and helping students formulate better goals did not by itself lead them to attain those goals (Martins van Jaarsveld et al., 2025b). If the benchmark is chosen by the student, the variation between students that matters most may lie less in the quality of their intentions than in how the intentions are carried out.

Most evidence on academic procrastination comes from variable-centered research that treats it as a relatively stable disposition and relates it to outcomes (Gustavson & Miyake, 2017; Steel & Klingsieck, 2016). In learning analytics, lateness is usually summarized as a rate or a count (Cormack et al., 2020; You, 2015). Both approaches implicitly assume that students differ in how much they delay, not in how their delay develops over time. Yet a student who is somewhat late throughout a course and a student who starts punctually and falls further behind with every assignment can have the same average lateness while facing very different problems. Temporal analyses of submission behavior have begun to address this (Conde et al., 2024), but trajectories of timeliness against deadlines that students set themselves have rarely been modeled.

If students differ in how their timeliness develops, the next question is where the difference originates. One possibility is that students who drift behind intended less from the start, aiming for lower grades or making less complete and less coherent plans. Another is that students with similar intentions and similar plans differ in enacting them. The two explanations call for different kinds of support: the first for help with goal setting and planning, the second for help at the moments when plans have to be carried out. Distinguishing them requires measures of intention and planning that were fixed before the trajectories unfolded, so that they are not themselves products of the behavior they are meant to explain.

We examine these questions in an asynchronous course at a Finnish university in which students chose a target grade and declared a personal deadline for each required assignment before the first assignment was due. Using the submission records of four cohorts, we identify latent trajectories of timeliness against these personal deadlines with growth mixture modeling, describe the temporal organization of being on time within each trajectory with sequence analysis, and test whether the trajectories are associated with the target grades and plans that students set at the start. The study contributes to research on individual differences in self-regulated learning in three ways. First, it treats timeliness as a trajectory within a course rather than as a trait score or an aggregate rate. Second, it combines two complementary person-centered methods, one to identify trajectories and one to describe their stability. Third, by separating variables fixed before the trajectories from variables derived from them, it tests whether differences in timeliness reflect differences in intention and planning or in enactment.

2 Theoretical background

2.1 Self-regulated learning and time management in flexible higher education

Models of self-regulated learning describe learning as a cyclical process in which learners set goals and plan their work (forethought), carry out and monitor the task (performance), and evaluate what they have achieved (self-reflection) (Zimmerman, 2000, 2002). Regulation in this view is goal-directed: learners monitor and control their cognition, motivation and behavior in relation to standards they have adopted (Pintrich, 2004). Time management runs through all three phases. It begins with deciding when to do what, continues with keeping to that decision while other demands compete for the same time, and ends with judging whether the plan was realistic.

In online and blended higher education, time management and effort regulation are among the strategies most consistently associated with achievement (Broadbent, 2017; Broadbent & Poon, 2015), and instruments for measuring self-regulation in online learning treat the setting of realistic

deadlines as part of the construct (Broadbent et al., 2022). Flexible course designs remove many of the external cues, such as scheduled sessions and common deadlines, that otherwise pace students' work, so that structure has to be generated by students themselves (Wong et al., 2019). Where it is not, postponed work can accumulate unnoticed until it surfaces as late or missing submissions.

2.2 From setting deadlines to keeping them

Goal-setting theory holds that specific goals direct attention and effort, and that commitment to a goal is a precondition for its effect on performance (Locke & Latham, 2006). Self-imposed deadlines can be understood as a form of such commitment: by choosing a date in advance, students bind their future selves to a course of action. Experimental work shows that people use self-imposed deadlines to counter their own tendency to delay, even though the deadlines they choose are not always the ones that would serve them best (Ariely & Wertenbroch, 2002).

Setting a goal and pursuing it are nonetheless distinct processes. A systematic review of goal-setting prompts in higher education maps how, why and when students are asked to set goals (Martins van Jaarsveld et al., 2025a), and a study in which a conversational agent guided students' goal setting found that guidance improved the quality of the goals but not their attainment (Martins van Jaarsveld et al., 2025b). In a similar vein, course deadlines did not prevent students from concentrating their learning shortly before exams; strategy use depended on individual differences as well as on course design (Theobald et al., 2021).

These findings point to a gap between intention and enactment. We use the term enactment gap for the difference between what students commit to at the start of a course and what they do when the committed moment arrives. Self-set deadlines make this gap observable. Because each student's timeliness is measured against a deadline the student chose, lateness indexes the enactment of the student's own plan rather than compliance with an instructor's timetable, and differences in lateness cannot be attributed to a timetable that was more demanding for some students than for others.

2.3 Academic procrastination: trait, process and trajectory

Academic procrastination, the voluntary delay of intended study activities despite expecting to be worse off for the delay, is common among students and has been linked to a range of antecedents, including low self-efficacy and task aversiveness (Steel & Klingsieck, 2016). Low self-efficacy for self-regulation in particular predicts procrastination among undergraduates (Klassen et al., 2008). Procrastination also predicts the attainment of academic goals over and above personality and motivation (Gustavson & Miyake, 2017), which makes it a meaningful individual difference in its own right.

Not all delay is dysfunctional, however. In a critical review of the Active Procrastination Scale, Pinxten et al. (2019) argue for distinguishing purposeful delay, in which students deliberately postpone work while remaining in control of it, from the passive delay usually meant by procrastination. For the present study, this distinction implies that students who are consistently late but still complete their work are not necessarily failing to regulate; their lateness may be a stable and deliberate pattern. Submission records show when work was delayed but not whether the delay was voluntary and irrational, as definitions of procrastination require (Steel & Klingsieck, 2016). We therefore use procrastination for the construct and describe the observed behavior as lateness relative to a personal deadline.

Procrastination also has dynamics. In a semester-long study, procrastination and self-efficacy influenced each other over time, forming vicious circles in which delay lowered self-efficacy and further increased delay, as well as virtuous circles in the opposite direction (Wäschle et al., 2014). Motivational regulation likewise shows interindividual differences in how it develops over the course of study (Kryshko et al., 2026). If delay feeds on itself, some students should show trajectories in which lateness escalates over a course while others remain stable. Detecting such patterns requires an approach that is both longitudinal and person-centered.

2.4 Regulation as a triggered process

Recent work conceptualizes regulation as a response to triggers. The Trigger Regulation Framework (TRF) (Järvelä & Hadwin, 2024) distinguishes a trigger situation, which can be described with contextual data such as timing and prior events; a trigger event, in which the learner detects a need to regulate; and a regulatory response, which becomes visible in trace data as a single action, a sequence or a pattern and can be adaptive or maladaptive.

TRF offers a way to read submission timing as regulation. An approaching personal deadline is a trigger situation that recurs throughout the course. Whether and when the student submits is trace data on the regulatory response. The trigger event, the student's noticing that a personal deadline is approaching or has passed, is not observable in our data and may be absent for some students, for example those who no longer follow the course. A trajectory of timeliness across the course is then a pattern of responses to a repeated trigger situation, and trajectory classes are profiles of such patterns. A trajectory in which delay grows can be read as a response that becomes increasingly maladaptive as the course progresses.

Trace data record what learners did, not what they intended to do, and interpreting traces as regulation requires knowing the purpose against which they were produced (Lämsä et al., 2025; Lämsä & Järvelä, 2026). In the present design this purpose is supplied by the students themselves, because each submission is read against a deadline and a target grade the student set. TRF has so far been illustrated mainly in supervised settings in which the trigger situations arise within the task (Lämsä & Järvelä, 2026). In a flexible course taken by students who work alongside their studies, the trigger situations that disrupt a plan are more likely to be external to the task, such as competing deadlines or work commitments, and are not visible in log data. We therefore describe regulatory responses without claiming to observe what triggered them.

2.5 Capturing timeliness with trace data

Learning analytics research has moved from counting activities and predicting outcomes toward modeling learning processes (Gašević et al., 2015; Saqr, López-Pernas, Helske, et al., 2024). Person-centered longitudinal methods are central to this shift because they allow for the possibility that groups of students follow qualitatively different courses of development. Growth mixture models identify latent classes of individuals who share a trajectory over time and estimate the shape of each class's trajectory (Proust-Lima et al., 2017). In the course studied here, earlier person-centered work that combined trace data with repeated self-reports identified distinct longitudinal patterns in how students' engagement and self-regulation developed (Heikkinen et al., 2025). Applied to timeliness, growth mixture models can show whether students differ in their level of lateness, in how lateness develops, or in both.

Sequence analysis complements growth modeling. It treats each student's course as an ordered sequence of states and describes how states follow each other, how often they change and how long they last (Gabadinho et al., 2011; Saqr, López-Pernas, Helske, et al., 2024). In education it has been used to uncover time management strategies from trace data (Uzir et al., 2019), to study procrastination as the temporal dynamics of being on time in a programming course (Conde et al., 2024), to follow students' online tactics and strategies across courses (Saqr et al., 2023) and to identify multi-channel learning trajectories of performance and engagement that remained largely stable across a semester (Gao et al., 2025). Where a growth model summarizes a trajectory by its level and slope, sequence analysis captures stability, volatility and order, features that can separate students whose average trajectories look alike.

2.6 The present study

The present study examines submission timeliness in an asynchronous course in which students chose a target grade and declared a personal deadline for each required assignment before the first assignment was due. Timeliness was measured against these personal deadlines across up to eleven assignments. We address three research questions:

RQ1. What latent trajectories of submission timeliness, measured against students' self-set deadlines, can be identified across a course?

RQ2. How do the trajectory classes differ in the temporal organization of being on time, that is, in the stability and transitions of their submission sequences?

RQ3. Are the trajectory classes associated with what students intended and planned at the start of the course (target grade, plan completeness and coherence)?

3 Method

3.1 Educational context and course design

The study was conducted in an asynchronous online course in project management for first-year business students at a Finnish university of applied sciences. The course ran in the Moodle learning management system and lasted 18-20 weeks depending on the length of the semester. At the start of the course, students chose a target grade on a scale from 1 to 5. The course comprised 11 assignments, and the target grade determined how many of them a student was required to complete: 4 to grade 1, 6 for grade 2, 8 for grade 3, 10 for grade 4, and all 11 for grade 5. As their first task, students submitted a goals-and-schedule form in which they declared a personal deadline for each required assignment. Lateness in this course is therefore defined against each student's own commitment rather than against an instructor-imposed schedule.

Assignments were assessed as accepted or returned for completion. In practice, a submitted assignment was accepted, and lateness relative to the personal deadline did not affect assessment. The final grade (0 to 5, with 0 denoting a fail) therefore reflected how many assignments a student completed, using the same thresholds as for the target grade; completing fewer than four assignments meant a fail. Students therefore could exceed or fall short of their target grade.

3.2 Participants and data

Data came from four consecutive cohorts (summer 2023, autumn 2023, spring 2024 and summer 2024). There were altogether 302 students enrolled for the courses. The analytic sample comprised the 262 students who submitted the goals-and-schedule form in which the personal deadlines were declared; all of them also had submission records in the learning management system. Of these, 131 were in the experimental and 127 in the control condition of the field experiment described below, and condition and final grade were not recorded for four students. The university gave a research permit for the study. Students gave informed consent, and the study followed the ethical principles for research with human participants of the Finnish National Board on Research Integrity (TENK, 2019).

This is the first study to use these data to analyze the timeliness of students' submissions with the aim of identifying distinct profiles of being late. The course also hosted a randomized field experiment on self-regulation support, and two earlier studies drew on the same data with different aims. The first examined how an early survey-based intervention and a later notification to students who had fallen behind their own schedule shaped students' self-reported self-regulation and the transition networks of their learning activities (Heikkinen et al., 2026a). The second used sequence analysis of learning sessions to compare the learning behavior of notified and non-notified students after the notification (Heikkinen et al., 2026b). In both studies, lateness served only to trigger the notification and to form the comparison groups; neither modeled timeliness itself or differences between students in how it developed over the course. In the experimental condition, students received feedback prompts on self-regulation based on an early survey and, when they fell behind the schedule they had declared, an email notification asking them to revise it; students in the control condition received the standard course communications (Appendix B). The interventions are not evaluated here, but the distribution of the conditions across the trajectory classes is reported in Section 4.5 and Appendix B.

3.3 Measures

3.3.1 Personal deadlines

Deadlines were taken from the goals-and-schedule form and set to 23:59:59 Helsinki time on the declared day. When a student had submitted the form more than once (19 students), the earliest form giving a usable date was retained. Declared dates outside a window from 30 days before to 400 days after the form submission (37 dates) were treated as invalid.

3.3.2 Submission timeliness

For each student and required assignment, timeliness was computed as the number of days between the submission timestamp and the personal deadline, with negative values indicating submission ahead of the deadline. When a student submitted work without having declared a deadline for it (189 assignments), timeliness was computed against the median deadline declared by the student's cohort for that assignment. This benchmark derives from other students' plans and is therefore independent of the submission being scored. A benchmark based on the cohort's median submission time was rejected because it is derived from the same behavior that is being measured; results with that benchmark are reported in Section 4.5.

Required assignments that a student never submitted (611 of 2,402 required student-assignment pairs) were entered at a ceiling value of 102 days equivalent with the hard deadline of the course, after which no submissions were accepted, as were the 18 submissions that arrived more than 95 days after the personal deadline. We treated non-submission as the most extreme value of timeliness rather than as missing data, for four reasons. First, each required assignment was selected by the student through the target grade and scheduled by the student in the goals-and-schedule form; not submitting it is the most extreme outcome of the same regulatory process that produces lateness, not an absence of information. Second, omitting non-submissions would estimate the trajectories from submitted work only, so that the classes with the most delay would be represented by their remaining submissions and would appear more punctual than they were. Third, maximum likelihood estimation treats omitted observations as missing at random, an assumption that is implausible here because whether an assignment is submitted is directly related to the timeliness being modeled. Fourth, the ceiling is tied to the data rather than chosen arbitrarily: lateness beyond 95 days was treated as implausible due to the hard deadline on the course, and the ceiling was set one week beyond this limit, so that a non-submission always scores later than any plausible submission while remaining close to the observed range.

3.3.3 Being-on-time states

Following Conde et al. (2024), each required assignment was coded as Early, On-time, Late or Missing relative to the personal deadline. An assignment was coded On-time when it was submitted on the day of the personal deadline, Early when it was submitted before that and Late when it was submitted after it. Because the number of required assignments depended on the target grade, sequences were of unequal length; positions beyond a student's own requirement were coded as void rather than missing.

3.3.4 Intentions and plans at course start

Target grade (1 to 5) was taken from the goals-and-schedule form. Plan quality described the declared deadlines themselves and was indexed by two primary indicators, *plan completeness* and the *logicality score*, and six secondary indicators. *Plan completeness* was the share of required assignments with a declared deadline. The *logicality score* (0 to 100) captured the chronological consistency of the declared deadlines. It was defined as $100 \times (1 - \min[1, v / (k - 1)])$, where k is the number of valid declared deadlines and v the number of violations: a deadline earlier than that of the preceding assignment, two consecutive assignments scheduled for the same day, or a deadline before the date on which the form was submitted. The score was computed for students who declared at least three deadlines (n = 235), and it was complemented by counts of chronology violations, backwards deadlines and same-day deadlines, the plan span in days, the mean gap between consecutive deadlines and the irregularity of the gaps (coefficient of variation). Both target grade and plan quality were fixed before the first assignment was due.

3.3.5 Grades

Final grade (1 to 5, 0 = fail) and grade deviance (final minus target grade) were taken from course records. They are used only to describe the classes (Section 3.4.3).

3.4 Data analysis

3.4.1 Growth mixture modeling

Timeliness trajectories were estimated with latent class mixed models using the hlme function of the lcmm package (Proust-Lima et al., 2017) in R 4.5.2 (R Core Team, 2025). Submissions ahead of the deadline were set to zero and the values were standardized before estimation, so that the model distinguishes degrees of lateness but not degrees of earliness; earliness is described through the being-on-time states (Section 3.3.3). Furthermore, our aim is to capture delay relative to plan: a student who submits two days early and one who submits thirty days early have both kept the plan, and because personal deadlines could lie weeks ahead, unbounded earliness would otherwise dominate the variance of the first assignments. Negative values are therefore used only in the observed trajectories (Figure 1) and in the being-on-time states. For the same reason, the class that stays at or ahead of its deadlines is labeled *On track* rather than early: the model identifies the absence of lateness, not the degree of lateness. Assignment position was normalized to a scale from 0 to 1 within each student, so that trajectories were comparable regardless of how many assignments the target grade required. The model included a random intercept, class-specific intercepts and linear slopes, and 100 random starts per solution. Class labels were assigned from the fitted profiles (first by the share of required assignments never submitted, then by mean lateness), so that labels followed the classes when they were renumbered across refits.

We specified a linear growth term for three reasons. First, the aim was to distinguish the direction of change across the course, stable versus increasing delay, for which a linear slope is the most parsimonious summary. Second, students contributed between four and eleven time points depending on their target grade, and curvilinear class-specific terms would add parameters and increase the risk of unstable or locally optimal solutions in a sample of this size. Third, the apparent bending of the observed class means in later assignments (Figure 1) is computed on the students who were still submitting at each deadline and therefore partly reflects changing class composition rather than curvature within students.

Solutions with one to six classes were estimated and compared on the Bayesian information criterion (BIC; Nylund et al., 2007), entropy, mean posterior probabilities and smallest class size. Because BIC did not reach a minimum within this range, the number of classes was decided on interpretability: we retained the most parsimonious solution in which every class represented a qualitatively distinct trajectory. Fit statistics for all solutions are reported in Table 2.

3.4.2 Sequence analysis

Being-on-time sequences were analyzed with the TraMineR package (Gabadinho et al., 2011). Dissimilarities between sequences were computed with optimal matching, using substitution costs derived from the observed transition rates (Studer & Ritschard, 2016), and rows in the index plots were ordered by Ward clustering of these dissimilarities. For each student we computed the proportion of assignments in each state, the longest unbroken runs of Late and Missing states and the positions of the first Late and first Missing state. Four indices of sequence stability were computed with the codyna package (Tikka & Saqr, 2026): the transition rate, the share of consecutive assignments in which the state changed; the self-loop tendency, the share of transitions in which it stayed the same; longitudinal entropy, the Shannon entropy of the student's state distribution normalized to the range 0 to 1; an integrative potential, the share of assignments in a favorable state

(here Early or On-time), weighted by position so that later assignments count more (linear weights). Transitions between states were modeled for each class with transition network analysis using the tna package (Tikka et al., 2025): each class network consists of the first-order transition probabilities between the four states and the probabilities of the initial state, and edges with probabilities below .05 were pruned.

3.4.3 Comparing the classes without circular inference

Class comparisons were organized by the relation of each variable to the data used to estimate the classes (Table 1). Timeliness and all sequence indicators are derived from the same submission records that define class membership, so that differences between classes on these variables are expected by construction. We report them descriptively to characterize the classes. Sequence indicators describe how stable being on time was within each class.

Target grade and plan quality were fixed before the trajectories unfolded and did not enter the model. These variables were compared across classes with Kruskal-Wallis tests, with epsilon-squared (ε^2) as the effect size. Three indicators were specified as primary and are reported in Table 5: target grade (intention), plan completeness and the logicality score (plan quality). The six remaining plan indicators are secondary and are reported in Appendix A. The Benjamini-Hochberg procedure (Benjamini & Hochberg, 1995) was applied separately within the primary and the secondary set; both unadjusted and adjusted p values are reported, and Dunn's pairwise tests with Holm adjustment were interpreted only for indicators with a significant adjusted omnibus test. Target grade determines the number of required assignments and thus the length of each student's sequence.

Comparisons used the most likely class for each student. Given the high classification quality of the retained solution (entropy = .931), bias from ignoring classification uncertainty was expected to be small (Asparouhov & Muthén, 2014). Robustness was examined by comparing the share of assignments scored against an imputed deadline across classes (Kruskal-Wallis test) and by testing the association between class membership and experimental condition (χ^2 test with Cramér's V).

Table 1 *Variables by their relation to the data used to estimate the classes*

Role	Variables	Treatment
Model input and derived from it	Days relative to the personal deadline, including non-submissions; being-on-time states; proportion of states, sequence indices (transition rate, self-loop tendency, longitudinal entropy, integrative potential), transition networks, run lengths, first Late and first Missing positions	Descriptive only
Fixed before the trajectories	Target grade; plan quality indicators	Kruskal-Wallis, ε^2, Dunn (Holm)
Follow from the number of submissions	Final grade; grade deviance	Descriptive only
Robustness	Share of assignments scored against an imputed deadline; experimental condition	Kruskal-Wallis; χ^2

3.4.4 Software and data availability

All analyses were conducted in R 4.5.2 (R Core Team, 2025) with the packages lcmm 2.2.2 (Proust-Lima et al., 2017), TraMineR 2.2-13 (Gabadinho et al., 2011), tna 1.2.3 (Tikka et al., 2025), codyna 0.1.0 (Tikka & Saqr, 2026), tidyverse 2.0.0 (Wickham et al., 2019) and ggalluvial 0.12.6 (Brunson,

2020). The data are not publicly available because the consent given by the participants does not cover data sharing.

4 Results

4.1 Number of classes (RQ1)

All solutions with one to six classes converged, and BIC decreased across the whole range (Table 2). We retained four classes as the most parsimonious solution in which each class showed a qualitatively distinct trajectory; the five-class solution divided the Escalating delay class into a milder and a more severe variant of the same rising trajectory, and the six-class solution added a small mixed class (5.3%) of moderately rising lateness drawn from three of the four classes. Classification quality was high (entropy = .931; mean posterior probabilities .89 to .98), and the smallest class comprised 26 students (9.9%).

Table 2 *Fit statistics for latent class growth solutions with one to six classes*

Classes	Converged	BIC	ΔBIC	Entropy	Smallest class
1	yes	5051.0		1.000	100.0%
2	yes	4526.3	−524.7	.883	30.9%
3	yes	4354.5	−171.9	.917	22.1%
4 (retained)	**yes**	**4327.0**	**−27.5**	**.931**	**9.9%**
5	yes	4286.9	−40.0	.927	10.3%
6	yes	4275.6	−11.4	.920	5.3%

Note. ΔBIC relative to the solution with one class fewer. Log-likelihood (number of parameters) for one to six classes: −2514.4 (4), −2243.7 (7), −2149.4 (10), −2127.3 (13), −2098.9 (16) and −2084.9 (19). Mean posterior probability of the assigned class ranged from .956 to .972 (two classes), .940 to .976 (three), .888 to .982 (four), .873 to .989 (five) and .838 to .977 (six). Class sizes: five classes 121, 50, 35, 29 and 27; six classes 120, 49, 28, 28, 23 and 14.

4.2 Four trajectory classes (RQ1)

Figure 1 shows the observed mean timeliness of the four classes at each assignment, and Table 3 summarizes the classes. The largest class, On track (n = 124, 47%), stayed slightly ahead of its own deadlines at every assignment and submitted almost all of its required work. Persistently late (n = 26, 10%) was late from the first assignment, rose to about 65 days behind the personal deadline at the third assignment and then held a stable distance of about 40 days while continuing to submit. Escalating delay (n = 55, 21%) started level with the On track class, about eight days ahead of the deadline at the first assignment, crossed the deadline at the second and fell to about 38 days behind by the seventh; from the middle of the course, missing submissions accumulated. Withdrawing (n = 57, 22%) was the latest class at every deadline at which it appears, about 82 days behind at the third assignment, and most of its required work was never submitted.

The later parts of the Withdrawing and Escalating delay curves should be read with care. They are computed on the students who were still submitting, a number that falls quickly in the Withdrawing class (from 53 students at the first deadline to 5 at the fourth), so the apparent decline in lateness reflects survivorship rather than recovery. Grades followed the order of the classes (Table 3), but

because every submitted assignment was accepted, these differences describe what the trajectories amounted to rather than an independent outcome. Descriptively, Persistently late had the lowest mean target grade and a final grade close to it.

Table 3 *Trajectory classes: size, classification quality, signature and grades*

Class	n (%)	Posterior prob.	Signature	Final grade	Grade deviance
On track	124 (47.3)	.978	Flat just below zero across the course	4.35 (0.86)	0.52 (0.89)
Persistently late	26 (9.9)	.888	Late from the start; stable positive offset	3.04 (1.40)	−0.08 (0.84)
Escalating delay	55 (21.0)	.961	Starts punctual; delay grows over the course	1.70 (1.18)	−1.89 (1.14)
Withdrawing	57 (21.8)	.982	Very late from the start; submissions stop early	0.31 (0.64)	−3.41 (1.16)

Note. Grades are M (SD): final grade 0 to 5; grade deviance = final minus target grade. Grades are descriptive only, since they follow from the number of submitted assignments (Section 3.4.3). Posterior prob. = mean posterior probability of the assigned class.

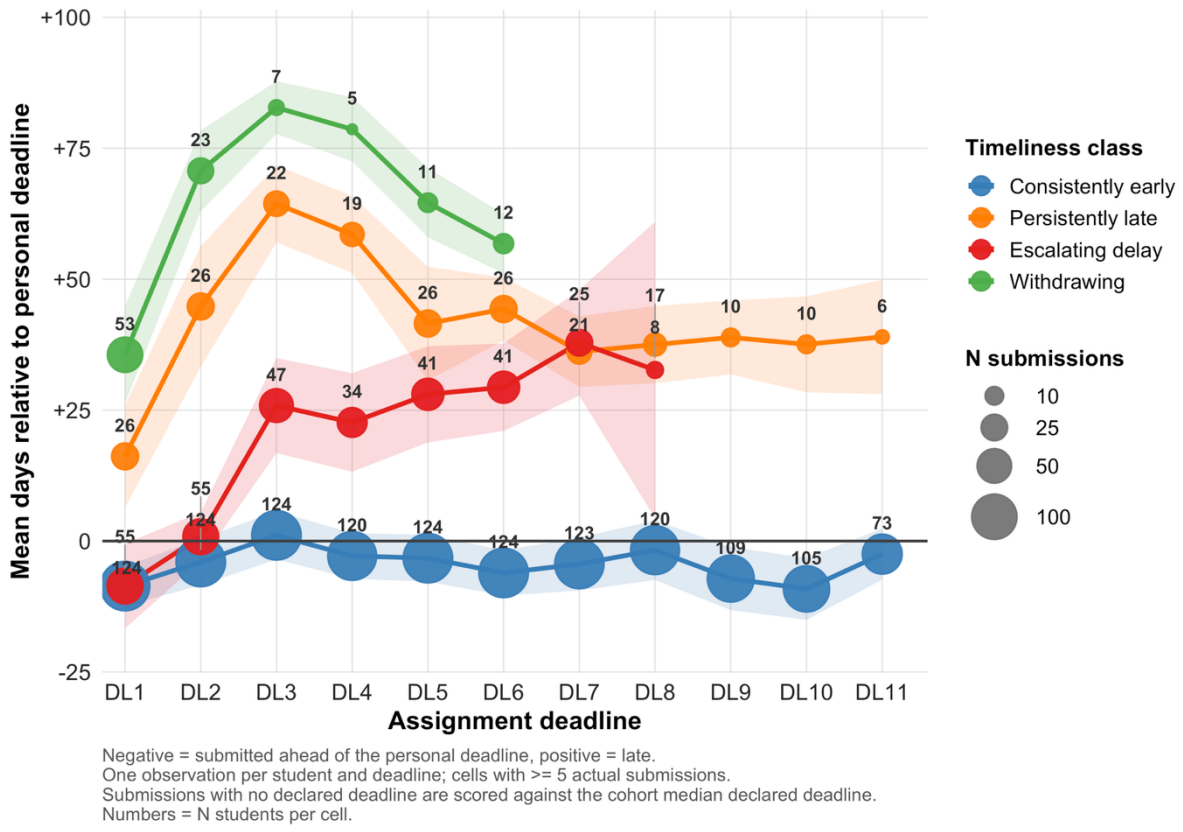


Figure 1 *Observed mean days relative to the personal deadline, by trajectory class and assignment.* Negative values indicate submission ahead of the student's own deadline. These values are retained here, whereas the model input set submissions ahead of deadline to zero (Section 3.4.1). Ribbons are 95% confidence intervals; point size and labels give the number of students per cell (minimum five). Submissions without a declared deadline are scored against the cohort median declared deadline. Declining lines in the later assignments of the Withdrawing and Escalating delay classes reflect the few students still submitting, not improvement. Students contribute only to the assignments

required at their target grade; DLs 1, 2, 5, and 6 for grade 1; DLs 1-3, and 5-7 for grade 2; DLs 1-8 for grade 3; DLs 1-10 for grade 4; and all DLs for grade 5.

4.3 Being-on-time sequences (RQ2)

Because the sequence states categorize the same timeliness values that define the classes, the differences between classes in the proportions of assignments in each state follow from how the classes were formed (Table 4). What the sequences add is information on how being on time developed over the course. Figure 2 shows the sequences of all students ordered by sequence similarity without reference to class membership. Four broad patterns are visible: sequences dominated by Early and On-time submissions, sequences mostly Late submissions, sequences that move from Late to Missing, and sequences in which Missing begin from the second or third deadline.

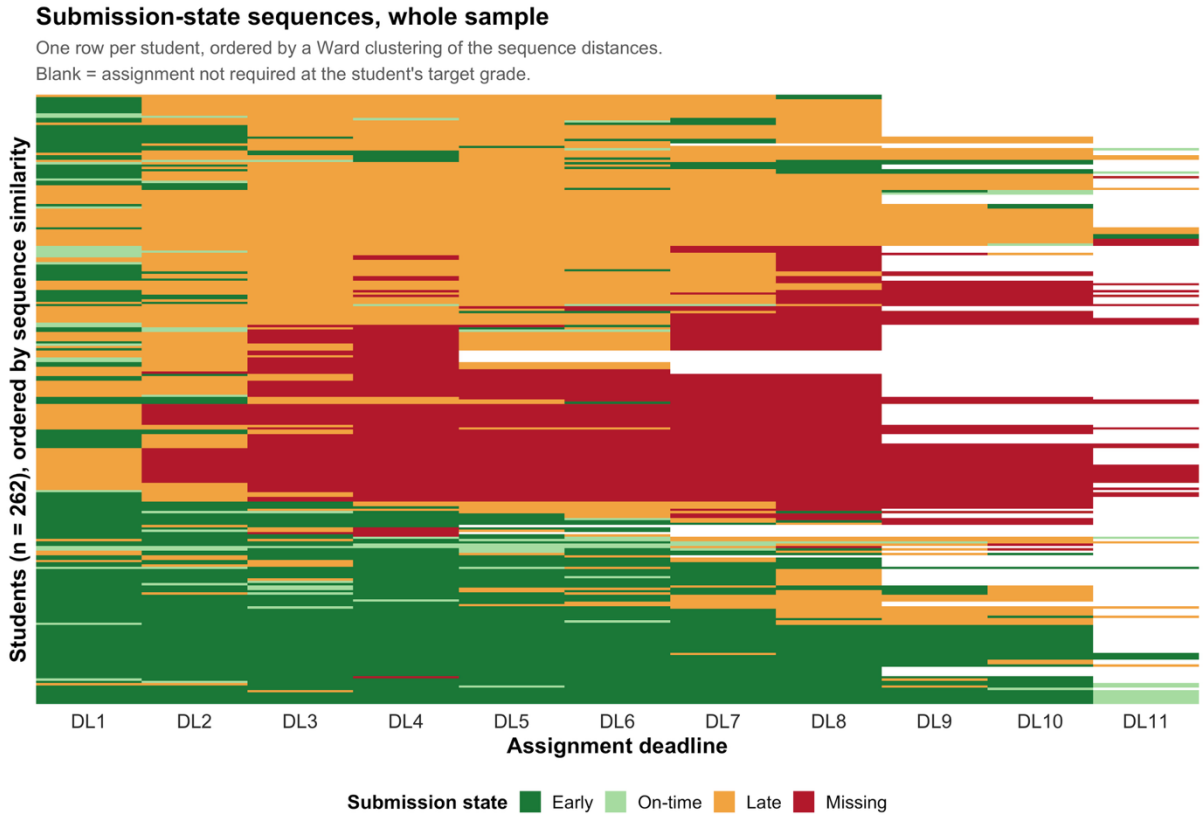


Figure 2 *Being-on-time sequences of all students (n = 262).* One row per student, ordered; blank cells mark assignments not required at the student's target grade.

The two classes with the most late submissions differed qualitatively (Table 4, Figure 3). Escalating delay had the highest transition rate (M = 0.325) and longitudinal entropy (M = 0.558), whereas Persistently late had the lowest longitudinal entropy (0.249), a low transition rate (0.189), a high self-loop tendency (0.811) and the longest unbroken runs of Late submissions (M = 6.58). The transition networks show the same contrast. In Persistently late, Late was almost absorbing (Late to Late .91), and Early submissions were mostly followed by Late ones (.80). In Escalating delay, an Early submission was followed by Early submission in about half of the cases (.52) and by a Late one in .39, Late led to Missing in .29 of the transitions, and once a submission was missing, the next one was missing too in .91. In Withdrawing, Late was followed by Missing more often than by Late (.54 versus .46), and Missing was virtually absorbing (.98). On track was the only class whose sequences converged on the favorable states (integrative potential M = 0.539, compared with 0.124 in

Escalating delay, 0.024 in Persistently late and 0.005 in Withdrawing): Early was followed by Early in .78 of the transitions, and a Late submission was followed by an Early one in .14. One late-submitting class was thus reliably late, whereas the other became increasingly unstable and drifted toward non-submission. The same transitions are shown for the whole sample as an alluvial diagram in Supplementary Material S1.

Table 4 *Sequence indicators by trajectory class (descriptive)*

Indicator	OT (124)	PL (26)	ED (55)	W (57)
% Early	53.88 (34.00)	5.48 (10.58)	17.82 (19.32)	1.78 (4.26)
% On-time	4.91 (8.28)	1.35 (5.21)	5.93 (12.81)	0.39 (2.10)
% Late	40.25 (33.73)	79.62 (24.02)	38.99 (22.32)	23.11 (18.24)
% Missing	0.97 (3.40)	13.55 (17.09)	37.25 (19.51)	74.72 (18.67)
Longitudinal entropy	0.314 (0.204)	0.249 (0.192)	0.558 (0.135)	0.321 (0.110)
Transition rate	0.244 (0.216)	0.189 (0.183)	0.325 (0.172)	0.185 (0.114)
Self-loop tendency	0.756 (0.216)	0.811 (0.183)	0.675 (0.172)	0.815 (0.114)
Integrative potential	0.539 (0.357)	0.024 (0.060)	0.124 (0.160)	0.005 (0.010)
Longest Late run	3.37 (3.22)	6.58 (3.09)	3.20 (2.14)	1.82 (1.50)
Longest Missing run	0.08 (0.27)	0.77 (0.86)	3.09 (1.84)	6.75 (2.55)
First Late position	3.91 (2.94)	1.36 (0.49)	2.89 (1.87)	1.19 (0.40)
First Missing position	7.70 (3.30)	6.00 (3.00)	5.96 (2.13)	3.00 (1.51)

Note. Cells are M (SD). Transition rate, self-loop tendency, longitudinal entropy and integrative potential were computed with codyna package (Tikka & Saqr, 2026); Early and On-time were favorable states for integrative potential. OT = On track; PL = Persistently late; ED = Escalating delay; W = Withdrawing. No significance tests are reported because all indicators derive from the data used to estimate the classes (Section 3.4.3). First-position indicators are computed for students who had the state (first Late: n = 98, 25, 54 and 57; first Missing: n = 10, 13, 52 and 57 for OT, PL, ED and W).

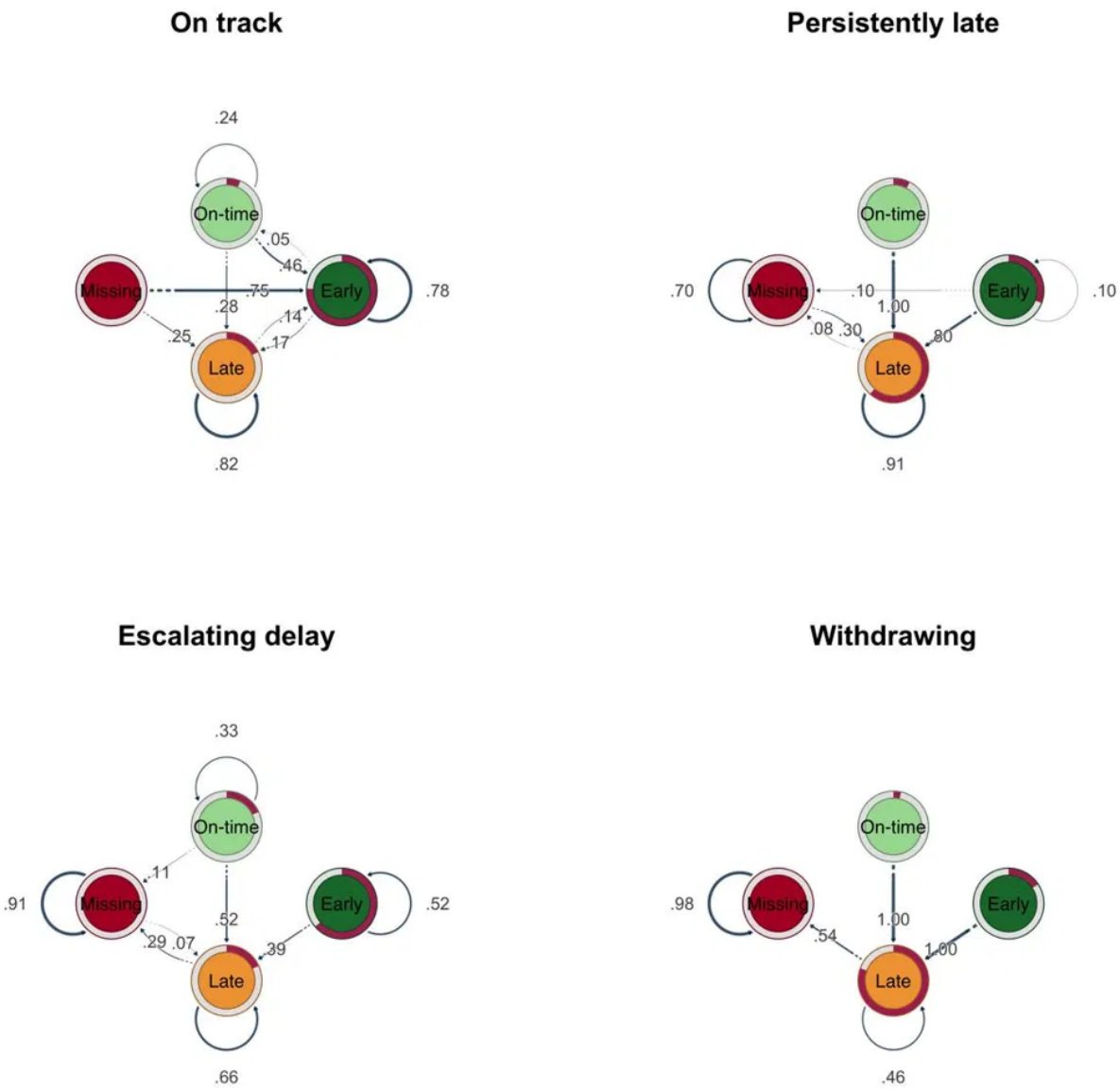


Figure 3 Transition networks of being-on-time states by trajectory class. *Nodes are four states; arrows give the probability that a submission in one state is followed by a submission in another state (or the same state) at the next required assignment, and edges with probabilities below .05 are not shown. The shaded rim on each node shows the probability of starting in that state. Probabilities out of rarely visited states, such as Missing in On track, rest on few transitions.*

4.4 Intentions and plans at course start (RQ3)

None of the three primary indicators differed across classes after adjustment (Table 5). Target grade showed a small effect that did not remain significant after adjustment ($H = 9.8$, unadjusted $p = .020$, adjusted $p = .060$, $\varepsilon^2 = .026$); descriptively, the mean target grade was highest in On track ($M = 3.83$) and lowest in Persistently late ($M = 3.12$). Plan completeness ($H = 1.8$, adjusted $p = .604$) and the logicality score ($H = 4.2$, adjusted $p = .366$) did not differ. Among the six secondary indicators (Appendix A), four did not differ. The two exceptions, plan span (adjusted $p = .004$, $\varepsilon^2 = .059$) and the

mean gap between consecutive deadlines (adjusted p = .034, ε^2 = .034), separated only On track from Escalating delay, whose plans were more compressed in time.

Table 5 *Target grade and plan quality at course start by trajectory class*

Variable	OT (124)	PL (26)	ED (55)	W (57)	H	p (unadj. / adj.)	ε^2
Target grade (1 to 5)	3.83 (0.87)	3.12 (1.11)	3.58 (0.83)	3.74 (0.97)	9.8	.020 / .060	.026
Plan completeness (%)	90.11 (23.36)	88.78 (23.03)	87.37 (23.81)	82.52 (32.11)	1.8	.604 / .604	−.004
Logicality score (0 to 100)	88.15 (24.58)	84.96 (21.34)	77.05 (33.44)	88.45 (21.08)	4.2	.244 / .366	.005

Note. Cells are M (SD). Kruskal-Wallis tests, df = 3; p values are shown unadjusted and Benjamini-Hochberg adjusted within the three primary indicators. OT = On track; PL = Persistently late; ED = Escalating delay; W = Withdrawing. Further plan quality indicators in Appendix A.

4.5 Robustness

The share of assignments scored against an imputed deadline did not differ across classes (H = 6.4, p = .094, ε^2 = .013) and was below 10% in every class, so the classes do not reflect the rule for undated submissions. Class membership was not significantly associated with experimental condition overall, $\chi^2(3) = 7.13$, p = .068, V = .17 (Appendix B, Table B.1). The share of experimental students was close to overall 51% in three classes (50% to 56%), whereas the Persistently late class contained fewer experimental students than expected (7 of 26; adjusted residual = -2.57). Because the notification in the experimental condition targeted students who had fallen behind their own schedule, this imbalance may reflect an effect of the intervention, which is examined in a companion study (Heikkinen et al., 2026b); the classes describe the pooled sample and are not used here to evaluate the interventions. Scoring undated submissions against the cohort's median submission time instead of the median declared deadline did not change the class solution: class sizes were identical, 259 of the 262 students were assigned to the same class (adjusted Rand index .97), and relative entropy was .936.

5 Discussion

5.1 Planned alike, enacted apart

This study identified four trajectories of submission timeliness against self-set deadlines in an asynchronous course. About half of the students stayed slightly ahead of their own deadlines throughout. One in ten were late from the first assignment and remained at a stable distance behind their plan. One in five started as punctually as the first group and fell increasingly behind, and one in five were very late from the outset and soon stopped submitting. Sequence analysis showed that the two late-submitting classes differed not only in when they fell behind but in how stable their behavior was. The classes were hardly distinguishable by the target grades and plans students had set before the first assignment was due.

The last finding is central to our interpretation. Students who went on to follow very different trajectories aimed for similar grades and produced plans of similar completeness and coherence. Differences in timeliness therefore did not originate in differences in intention or planning as recorded at the start of the course; they emerged as the plans were enacted. This is consistent with

evidence that improving the quality of goals does not by itself improve their attainment (Martins van Jaarsveld et al., 2025b) and that deadlines alone do not prevent postponement (Theobald et al., 2021). Moreover, with increasing discrepancy between goal setting and actual attainment, the students experience more negative and less positive affect (Theobald & Bellhäuser, 2021). It suggests that in flexible courses the decisive regulatory work happens less when students plan than when the planned moment arrives.

Two qualifications apply. First, the absence of significant differences in target grade and plan quality is not evidence that the classes were equivalent in intention and planning. Our claim is limited to what the design supports: the classes are not accounted for by the intentions and plans that students recorded at the start. Second, intention and planning were not entirely irrelevant. Persistently late students had the lowest mean target grade, although this difference did not remain significant after adjustment, and Escalating delay students made plans that were more compressed in time. The latter difference is small, but it may be an early signal of a plan with too little slack, which future studies could examine directly.

5.2 Two kinds of risk: stable lateness and escalating instability

The Persistently late class was late from the first deadline, showed the most stable sequences of all classes and continued to submit work. It also had the lowest target grades, and its final grades were close to those targets. Two readings are possible. The pattern may reflect purposeful delay (Pinxten et al., 2019): students who deliberately work on a later schedule than the one they declared, perhaps having declared deadlines they did not treat as binding. Or it may reflect calibrated, modest ambition: students who set lower goals and meet them on their own timetable. Both readings describe a stable pattern rather than a spiral, and whether this group needs support at all is an open question. Because grades in this course follow from the number of submissions, the closeness of final and target grades is not evidence of successful regulation.

The Escalating delay class began indistinguishable from the punctual class, drifted behind from the second deadline onwards and had the most volatile sequences, with missing work accumulating from the middle of the course. This pattern fits the vicious circle described by Wäschle et al. (2014), in which early slippage may lower self-efficacy and make further delay more likely. In terms of the Trigger Regulation Framework, it is a regulatory response that turns increasingly maladaptive over repeated trigger situations (Järvelä & Hadwin, 2024). Because these students start well and deteriorate, the timing of support seems most consequential for them.

The Withdrawing class was very late from the first deadline and missing from about the third. For these students the problem is less about keeping to a schedule than about staying in the course at all, and support probably has to address engagement or life circumstances rather than scheduling. Taken together, the three classes at risk show why a single lateness rate is insufficient: it would rank the classes by how much they delayed but would not reveal that one pattern is stable, one deteriorating and one a withdrawal from the course.

5.3 Reading trajectories as regulation

Reading submission timing as regulation depends on knowing what the student was aiming for. In this study, the student-set deadlines supplied that purpose (Lämsä et al., 2025): the same submission date means something different against different plans, and this is what allows timeliness to be

interpreted as the enactment of a plan rather than as compliance with a rule. The Trigger Regulation Framework (Järvelä & Hadwin, 2024) then allows the trajectories to be read as profiles of responses to a trigger situation that recurs at every personal deadline.

What the traces cannot show is why a plan was not kept. In a university of applied sciences, where many students work alongside their studies, the triggers that disrupt a plan are likely to lie outside the task: work shifts, competing deadlines in other courses, family demands. Unlike the task-internal triggers in the supervised settings described by Lämsä and Järvelä (2026), such triggers leave no trace in the learning management system. Future research could capture them with brief event-based prompts at the moment a personal deadline passes without a submission, linking the observed response to the situation that produced it.

5.4 Implications for practice

The results suggest two principles for supporting students in flexible courses. The first concerns timing. The classes already looked different at the first few personal deadlines (Figure 1), which indicates that the early weeks of a course are when differences between students begin to open up. A study of programming students similarly found that learning trajectories stayed largely stable over a semester and that early learning patterns were important for final course success (Gao et al., 2025). Because the classes were estimated from data covering the whole course, this is an observation about the trajectories rather than evidence that students can be classified reliably after a few deadlines; testing early identification is a task for future work. The second principle concerns differentiation. Stable lateness, escalating delay and withdrawal are different problems, while a single reminder sent to every student who misses a deadline treats them as one.

The Trigger Regulation Framework adds that support should respond to a detected trigger rather than to a stable characteristic of the student (Järvelä & Hadwin, 2024). A missed personal deadline is such a trigger, and the type of trajectory a student is on could inform what the response looks like: a planning conversation for a student whose delay is escalating, a check on engagement for a student who is withdrawing, and perhaps no intervention for a student whose lateness is stable and whose work is completed. Companion studies in the same course have examined interventions of this kind (Heikkinen et al., 2026a, 2026b).

Turning such insights into practice is itself a challenge. Many learning analytics initiatives describe or predict student behavior without acting on the results. A recent expert study identified institutional enablers for moving from learning analytics to educational interventions and found that the institution's strategic orientation influences the other enablers (Svetec et al., 2026). Trajectory-based support of the kind suggested here requires exactly this: an institution that treats timeliness data as a basis for pedagogical action rather than as a monitoring record.

5.5 Limitations and future research

Several limitations qualify the findings. The data come from a single course at a single institution in one discipline, and the smallest class comprised only 26 students; replication in other course formats and institutions is needed. Treating non-submissions as ceiling values is a substantive decision about what a trajectory of timeliness means, and although we consider it the most defensible choice (Section 3.3.2), it gives non-submission considerable weight in the class solution. BIC did not reach a minimum within the range of one to six classes, so the number of classes rests on interpretability.

The linear growth term summarizes the direction of change well but smooths over the early rise of the Persistently late class between the first and third assignments, which is not a composition effect because all students of that class were still submitting.

The observed class means in later assignments are based on the students still submitting, and a small share of submissions (under 10% in each class) was scored against an imputed deadline. Because every submitted assignment was accepted, no measure of learning or work quality independent of the number of submissions was available, and the consequences of the trajectories could only be described through grades. Triggers external to the task, which are likely to matter most in this population, were not observable. Finally, the data come from a randomized field experiment; class membership was not significantly associated with condition overall, although the Persistently late class contained fewer experimental students than expected (Section 4.5); the interventions were not modeled, and the design does not support causal claims about the trajectories. Future work should replicate the trajectories in courses with graded assignments, where their relation to the quality of work can be tested, and examine whether trajectories can be identified from the first few deadlines and used to trigger differentiated support.

6 Conclusions

Students in a flexible course who set their own deadlines followed four distinct trajectories of timeliness: on track, persistently late, escalating delay and withdrawing. The trajectories were not accounted for by the target grades and plans students set at the start of the course. Students aimed and planned alike but differed in how they enacted their plans.

Combining growth mixture modeling with sequence analysis showed that the two late-submitting trajectories were different kinds of risk, one stable and one increasingly unstable, a distinction that an aggregate lateness rate merges. Methodologically, the study shows how classes derived from trace data can be characterized without circular inference, by describing the variables that formed them and testing explanations only with variables fixed beforehand.

For practice, the findings suggest attending to the first personal deadlines of a course and matching support to the type of delay, rather than treating all late students alike.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to structure the manuscript and draft parts of the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

Ariely, D., & Wertenbroch, K. (2002). Procrastination, deadlines, and performance: Self-control by precommitment. *Psychological Science, 13*(3), 219–224.

Asparouhov, T., & Muthén, B. (2014). Auxiliary variables in mixture modeling: Three-step approaches using Mplus. *Structural Equation Modeling, 21*(3), 329–341.

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological), 57*(1), 289–300.

Broadbent, J. (2017). Comparing online and blended learner's self-regulated learning strategies and academic performance. *The Internet and Higher Education, 33*, 24–32. https://doi.org/10.1016/j.iheduc.2017.01.004

Broadbent, J., Panadero, E., Lodge, J. M., & Fuller-Tyszkiewicz, M. (2023). The self-regulation for learning online (SRL-O) questionnaire. *Metacognition and Learning, 18*(1), 135–163. https://doi.org/10.1007/s11409-022-09322-x

Broadbent, J., & Poon, W. L. (2015). Self-regulated learning strategies & academic achievement in online higher education learning environments: A systematic review. *The Internet and Higher Education, 27*, 1–13.

Brunson, J. C. (2020). ggalluvial: Layered grammar for alluvial plots. *Journal of Open Source Software, 5*(49), 2017. https://doi.org/10.21105/joss.02017

Conde, J., López-Pernas, S., Barra, E., & Saqr, M. (2024). The temporal dynamics of procrastination and its impact on academic performance: The case of a task-oriented programming course. In *Proceedings of the 39th ACM/SIGAPP Symposium on Applied Computing* (pp. 48–55). https://doi.org/10.1145/3605098.3636072

Cormack, S. H., Eagle, L. A., & Davies, M. S. (2020). A large-scale test of the relationship between procrastination and performance using learning analytics. Assessment & Evaluation in Higher Education, 45(7), 1046–1059. https://doi.org/10.1080/02602938.2019.1705244

Finnish National Board on Research Integrity TENK. (2019). *The ethical principles of research with human participants and ethical review in the human sciences in Finland* (Publications of the Finnish National Board on Research Integrity TENK 3/2019).

Gabadinho, A., Ritschard, G., Müller, N. S., & Studer, M. (2011). Analyzing and visualizing state sequences in R with TraMineR. *Journal of Statistical Software, 40*(4), 1–37.

Gao, Z., Yan, H., Liu, J., Cui, C., Wang, J., Zhang, X., Sun, X., & Feng, J. (2025). Untangling complexity of competency evolution: Multi-channel learning trajectories of score, engagement and problem-solving efficiency. *British Journal of Educational Technology*. https://doi.org/10.1111/bjet.70017

Gašević, D., Dawson, S., & Siemens, G. (2015). Let's not forget: Learning analytics are about learning. *TechTrends, 59*(1), 64–71. https://doi.org/10.1007/s11528-014-0822-x

Gustavson, D. E., & Miyake, A. (2017). Academic procrastination and goal accomplishment: A combined experimental and individual differences investigation. *Learning and Individual Differences, 54*, 160–172. https://doi.org/10.1016/j.lindif.2017.01.010

Heikkinen, S., Saqr, M., Malmberg, J., & Tedre, M. (2025). A longitudinal study of interplay between student engagement and self-regulation. *International Journal of Educational Technology in Higher Education, 22*, Article 21. https://doi.org/10.1186/s41239-025-00523-3

Heikkinen, S., Bellhäuser, H., van der Stappen, E., & Saqr, M. (2026a). *Self-regulated learning interventions: Seeding effects and behavioral side-effects* [Preprint]. SSRN. https://doi.org/10.2139/ssrn.6280359

Heikkinen, S., Bellhäuser, H., van der Stappen, E., & Saqr, M. (2026b). Behind schedule: Capturing the effects and side-effects of an intervention on the learning process. In S. López-Pernas et al. (Eds.), *The changing landscapes of smart learning environments: Pedagogy, AI, collaborative intelligence, and ethics. Proceedings of the 9th International Conference on Smart Learning Environments* (in press). Springer.

Järvelä, S., & Hadwin, A. (2024). Triggers for self-regulated learning: A conceptual framework for advancing multimodal research about SRL. *Learning and Individual Differences, 115*, 102526. https://doi.org/10.1016/j.lindif.2024.102526

Klassen, R. M., Krawchuk, L. L., & Rajani, S. (2008). Academic procrastination of undergraduates: Low self-efficacy to self-regulate predicts higher levels of procrastination. *Contemporary Educational Psychology, 33*(4), 915–931.

Kryshko, O., Becker, M., Fleischer, J., & Leutner, D. (2026). Self-regulation of motivation in university students: A longitudinal study of interindividual differences and intraindividual trajectories. *Learning and Individual Differences, 125*, 102817.

Lämsä, J., & Järvelä, S. (2026). Augmenting, not outsourcing: Leveraging AI to support students' self-regulated learning. *Theory Into Practice*. https://doi.org/10.1080/00405841.2026.2721203

Lämsä, J., de Mooij, S., Aksela, O., Athavale, S., Bistolfi, I., Azevedo, R., Bannert, M., Gasevic, D., Molenaar, I., & Järvelä, S. (2025). Measuring secondary education students' self-regulated learning processes with digital trace data. *Learning and Individual Differences, 118*, 102625. https://doi.org/10.1016/j.lindif.2024.102625

Locke, E. A., & Latham, G. P. (2006). New directions in goal-setting theory. *Current Directions in Psychological Science, 15*(5), 265–268.

Martins van Jaarsveld, G., Wong, J., Baars, M., Specht, M., & Paas, F. (2025a). Goal setting in higher education: How, why, and when are students prompted to set goals? A systematic review. *Frontiers in Education, 9*, 1511605. https://doi.org/10.3389/feduc.2024.1511605

Martins van Jaarsveld, G., Wong, J., Baars, M., Specht, M., & Paas, F. (2025b). Scaling goal-setting interventions in higher education using a conversational agent: Examining the effectiveness of guidance and adaptive feedback. In *Proceedings of the 15th International Learning Analytics and Knowledge Conference* (pp. 328–338).

Nylund, K. L., Asparouhov, T., & Muthén, B. O. (2007). Deciding on the number of classes in latent class analysis and growth mixture modeling: A Monte Carlo simulation study. *Structural Equation Modeling, 14*(4), 535–569.

Pinxten, M., De Laet, T., Van Soom, C., Peeters, C., & Langie, G. (2019). Purposeful delay and academic achievement: A critical review of the Active Procrastination Scale. *Learning and Individual Differences, 73*, 42–51. https://doi.org/10.1016/j.lindif.2019.04.010

Pintrich, P. R. (2004). A conceptual framework for assessing motivation and self-regulated learning in college students. *Educational Psychology Review, 16*(4), 385–407.

Proust-Lima, C., Philipps, V., & Liquet, B. (2017). Estimation of extended mixed models using latent classes and latent processes: The R package lcmm. *Journal of Statistical Software, 78*(2), 1–56.

R Core Team. (2025). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. https://www.R-project.org/

Saqr, M., López-Pernas, S., Helske, S., Durand, M., Murphy, K., Studer, M., & Ritschard, G. (2024). Sequence analysis in education: Principles, technique, and tutorial with R. In M. Saqr & S. López-Pernas (Eds.), *Learning analytics methods and tutorials: A practical guide using R*. Springer. 321-354.

Saqr, M., López-Pernas, S., Jovanović, J., & Gašević, D. (2023). Intense, turbulent, or wallowing in the mire: A longitudinal study of cross-course online tactics, strategies, and trajectories. *The Internet and Higher Education*. 57(April 2023). https://doi.org/10.1016/j.iheduc.2022.100902

Steel, P., & Klingsieck, K. B. (2016). Academic procrastination: Psychological antecedents revisited. *Australian Psychologist, 51*(1), 36–46. https://doi.org/10.1111/ap.12173

Studer, M., & Ritschard, G. (2016). What matters in differences between life trajectories: A comparative review of sequence dissimilarity measures. *Journal of the Royal Statistical Society: Series A (Statistics in Society), 179*(2), 481–511.

Svetec, B., Divjak, B., & Kadoić, N. (2026). From learning analytics to educational interventions. *International Journal of Educational Technology in Higher Education, 23*, 45. https://doi.org/10.1186/s41239-026-00620-x

Theobald, M., & Bellhäuser, H. (2021). Mind the gap!. Zeitschrift für Entwicklungspsychologie und Pädagogische Psychologie. https://doi.org/10.1026/0049-8637/a000218

Theobald, M., Bellhäuser, H., & Imhof, M. (2021). Deadlines don't prevent cramming: Course instruction and individual differences predict learning strategy use and exam performance. Learning and Individual Differences, 87, 101994. https://doi.org/10.1016/j.lindif.2021.101994

Tikka, S., López-Pernas, S., & Saqr, M. (2025). tna: An R package for transition network analysis. Applied Psychological Measurement, 49(6), 326–328. https://doi.org/10.1177/01466216251348840

Tikka, S., & Saqr, M. (2026). codyna: Complex dynamic systems (R package version 0.1.0). https://doi.org/10.32614/CRAN.package.codyna

Uzir, N. A., Gašević, D., Matcha, W., Jovanovic, J., Pardo, A., Lim, L.-A., & Gentili, S. (2019). Discovering time management strategies in learning processes using process mining techniques. *Lecture Notes in Computer Science, 11722*, 555–569.

Wäschle, K., Allgaier, A., Lachner, A., Fink, S., & Nückles, M. (2014). Procrastination and self-efficacy: Tracing vicious and virtuous circles in self-regulated learning. *Learning and Instruction, 29*, 103–114. https://doi.org/10.1016/j.learninstruc.2013.09.005

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., ... Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software, 4*(43), 1686. https://doi.org/10.21105/joss.01686

Wong, J., Baars, M., Davis, D., Van Der Zee, T., Houben, G.-J., & Paas, F. (2019). Supporting self-regulated learning in online learning environments and MOOCs: A systematic review. *International Journal of Human-Computer Interaction, 35*(4–5), 356–373. https://doi.org/10.1080/10447318.2018.1543084

You, J. W. (2015). Examining the effect of academic procrastination on achievement using LMS data in e-learning. Educational Technology & Society, 18(3), 64–74.

Zimmerman, B. J. (2000). Attaining self-regulation: A social cognitive perspective. In M. Boekaerts, P. R. Pintrich, & M. Zeidner (Eds.), *Handbook of self-regulation* (pp. 13–39). Academic Press.

Zimmerman, B. J. (2002). Becoming a self-regulated learner: An overview. *Theory Into Practice, 41*(2), 64–70. https://doi.org/10.1207/s15430421tip4102_2

Appendix A. Plan quality by trajectory class

Table A.1 *Plan quality indicators by trajectory class*

Indicator	OT (124)	PL (26)	ED (55)	W (57)	H	p (unadj. / adj.)	ε^2
Plan completeness (%)	90.11 (23.36)	88.78 (23.03)	87.37 (23.81)	82.52 (32.11)	1.8	.604 / .604	−.004
Logicality score (0 to 100)	88.15 (24.58)	84.96 (21.34)	77.05 (33.44)	88.45 (21.08)	4.2	.244 / .366	.005
Chronology violations	0.93 (1.98)	0.92 (1.52)	1.76 (2.82)	0.84 (1.90)	3.3	.342 / .514	.001
Backwards deadlines	0.07 (0.26)	0.04 (0.20)	0.04 (0.19)	0.09 (0.29)	1.6	.650 / .650	−.005
Same-day deadlines	0.84 (1.96)	0.88 (1.53)	1.69 (2.77)	0.75 (1.87)	4.6	.203 / .407	.006
Plan span (days)	72.74 (26.71)	63.08 (24.90)	54.55 (31.77)	62.30 (25.50)	17.3	<.001 / .004	.059
Mean gap between deadlines (days)	9.79 (9.58)	10.48 (5.42)	7.14 (4.98)	8.22 (3.49)	11.1	.011 / .034	.034
Gap irregularity (CV)	0.58 (0.64)	0.63 (0.44)	1.20 (3.64)	0.73 (1.11)	1.8	.621 / .650	−.006

Note. Cells are M (SD). Kruskal-Wallis tests, df = 3; p values are shown unadjusted and Benjamini-Hochberg adjusted, plan completeness and the logicality score within the three primary indicators (Table 5) and the other six within the secondary set; negative ε^2 values are reported as computed. The logicality score is based on 235 students who declared at least three deadlines; same-day deadlines accounted for 263 of the 284 violations, backwards deadlines for 17 and deadlines before the form submission for 4. A complete plan (yes/no) was not associated with class, $\chi^2(3) = 1.42$, $p = .70$.

Appendix B. The field experiment

The course hosted a randomized field experiment on self-regulation support, reported in two companion studies (Heikkinen et al., 2026a, 2026b). Students were randomly assigned to an experimental (n = 131) or a control condition (n = 127); condition was not recorded for four students. Students in the experimental condition received two interventions. Early in the course, they received feedback prompts on self-regulation based on their responses to a survey. Later, a student who had fallen behind the schedule declared in the goals-and-schedule form received a personalized email notification asking the student to revise the schedule. Students in the control condition received the standard course communications. The interventions are not evaluated in the present study; Table B.1 reports how the two conditions were distributed across the trajectory classes.

Table B.1 *Experimental condition by trajectory class*

Class	Control (n)	Experimental (n)	Experimental (%)	Adjusted residual
On track	55	69	55.6	1.51
Persistently late	19	7	26.9	−2.57
Escalating delay	26	28	51.9	0.18
Withdrawing	27	27	50.0	−0.13
Total	127	131	50.8	

Note. Adjusted residuals refer to the experimental condition; residuals for the control condition have the opposite sign. $\chi^2(3) = 7.13$, $p = .068$, Cramér's V = .17; permutation test $p = .069$. No residual exceeded the Bonferroni-corrected critical value for the eight cells ($|z| = 2.73$). Without the Persistently late class, $\chi^2(2) = 0.55$, $p = .758$. Four students without a recorded condition are excluded.

Supplementary material

S1. Alluvial diagram of transitions between submission states.

